GeoFusionNet / GeoCorr: Project Status Report

Status as of 2026-10-09, 19:20 UTC · main @ e92a340 · Prepared for Gobika S

Summary. The frozen Stage 0 baseline (A0f) scores **0.491 ± 0.009** overall on full-res DTU (3 seeds × 22 scans). The geometric prior does not help in any of the ways tested: as an input (B1), as a loss (L_geo), or as a confidence-gated replacement. In every case the effect is within noise or limited to a few outlier scans. Only one change has clearly helped so far: more cost-volume capacity (B2). It reaches **0.414 / 0.418** on two seeds, against A0f's 0.479 on seed 2. The gain comes from completeness, and per-view depth metrics did not improve. The third B2 seed, the B2 no-L_geo ablation and a frozen mono-feature variant are running or queued now.

1. Ongoing work

Queue / job	State	ETA
B2 seeds + no-geo (PID 3390614) scripts/queue_b2_seeds_nogeo.sh · log runs/queue_b2_seeds_nogeo.log	B2_capacity seed0: **done** (overall 0.414) B2_nogeo seed2: **running**, base phase step 18,900 of 20,000; then 6k fine-tune, 1.5k final phase, diagnostics, probe, 1600×1152 eval B2_capacity seed1: queued (full chain)	Queue done ~2026-10-11 07:30 UTC
B2_mono_feat (PID 3457546) scripts/queue_b2_mono_feat.sh · patch fc8c805	Waiting for QUEUE_DONE above. Then seed 2: identity test, train from B2_nogeo_finetune seed2, diagnostics, probe, eval. Matched control: B2_nogeo seed2.	Starts ~2026-10-11 07:30 UTC
B2 3-seed aggregate	scripts/aggregate_b2_3seed.sh (c54dfc2). Runs only after QUEUE_DONE and writes runs/B2_capacity/aggregated_1600x1152.json.	After queue

Every queue step started after commit e92a340 (2026-10-09 ~17:10 UTC) logs a `[provenance] git commit …` line.

2. Headline numbers (DTU test, 22 scans, 1600×1152, official protocol; mm, lower is better)



Overall (mean of Acc and Comp, mm). Bars start at 0.

Model	Acc	Comp	Overall	Notes
A0 (pre-fix)	0.447	0.586	0.517	Superseded
A0f Stage 0 (3 seeds)	0.437 ± 0.003	0.545 ± 0.016	**0.491 ± 0.009**	Frozen baseline (std over seeds, ddof=0)
A0f seed 2	0.435	0.523	0.479	Reference for single-seed tests
CL-MVSNet, local 640×480 plys	0.446	0.431	0.438	CL-MVSNet point clouds generated locally at 640×480 from CL-MVSNet's released checkpoint (not released point clouds), scored with our evaluator. Published: 0.329. The 0.438 vs 0.329 gap mixes test resolution (640×480 here vs the 1184×1600 set in CL-MVSNet's test script), the evaluation crop, fusion thresholds and checkpoint differences; its cause is unconfirmed. Not a like-for-like comparison with our models.
B2_capacity seed 2	0.473	0.362	0.418	Better than A0f s2 on 17/22 scans
B2_capacity seed 0	0.470	0.358	0.414	New, 2026-10-09
B2_capacity seed 1	Not measured yet (queued)			

3. How we got here

3.1 Fixing the pipeline (2026-09-23 to 09-25)

- **Constant-depth failure:** early A0 runs had ~67 mm error. The cause was edge-aware smoothness applied to raw mm depth. Normalizing by per-image mean depth fixed it. A wrong score sign was ruled out.
- **Evaluation:** evaluation before 2026-09-23 had no fusion filtering and used a non-official metric. It now uses GeoMVSNet's open3d fusion (prob 0.3, 0.2 mm, 4 views) and the official DTUeval protocol, cross-checked against the reference. Full resolution is 1600×1152. **Numbers from before the fix are not comparable.**
- **Undertraining:** A0 saw ~40k samples, about 9% of the reference recipe. A CL-MVSNet-style LR fine-tune helped only ~5% (val fine median 1.75 → 1.65–1.70 mm), so the LR schedule does not explain the plateau.
- **Cost-volume fixes:** out-of-view masking plus trilinear regularizer upsampling cut sawtooth from 41% to 0.1%. They are now the default.

3.2 Failure analysis: scan48 and the outlier scans (11, 13, 48, 62, 77)

- The main failure is **photometric ambiguity**. On wrong pixels, GT is the global cost minimum on only 30%, a secondary minimum on 41%, and no minimum on 28%. Our photometric loss itself prefers the wrong depth.
- Robust losses (L0.5, min-view) and best-k view aggregation did not fix it. Widening the depth windows at test time made it worse.
- The CL-MVSNet consistency loss (ICC + SCC, ported with ×0.0163 weight calibration) was within noise on aggregate val. On scan48 it was worse at the final checkpoint (4.34 vs 3.45 mm). **Dropped**; A0f (fixes only) was frozen as Stage 0.

3.3 Geometric prior G (Depth-Anything-V2-S anchored to SIFT points)

Test	Result (seed 2, matched control)	Verdict
B1: G as input (zero-init GGF)	Val fine +0.012 mm, coverage −1.2 pt; scan48 −0.44 mm, but the control's own swing between checkpoints is 0.39 mm; competing dips unchanged (18.53 vs 18.42%)	No gain
No-L_geo vs control	Val fine +0.025 mm; scan48 −0.20 mm; dips +0.15 pt	Noise, so L_geo has no measurable effect
Prior quality (22 scans)	Median |G−GT| 5–69 mm vs model 0.8–4 mm. On ambiguous pixels G is nearer the true dip on only 52.4%, and within 4 mm on only 11.5%. 16/49 scan48 views have no prior.	About a coin flip as a dip selector
Confidence gates (thresholds frozen on val)	Only "prior disagrees with coarse depth by ≥ 99.3 mm" passes the pre-registered rule (test: 21.2% coverage, 86.7% correct). Partly mechanical.	Fixes gross outliers only
Gated fusion, A0f s2 (22 scans)	0.479 → 0.470. Almost all of the gain is from scan48 (−0.116) and scan77 (−0.114); the other 20 scans are slightly worse.	Small, concentrated
Gated fusion, B2 s2 (refit, 89.3 mm)	0.418 → 0.415. scan48 0.708 → 0.706.	Near no-op on B2

3.4 Capacity: B2_capacity

- Changes: FPN 32/96/192, 8-group correlation cost, 3D UNet regularizer. 2.03M params vs A0f's 121k. Same 20k + 6k + 1.5k training chain as A0f.
- Fused point clouds are much better (comp 0.36 vs 0.52; scan48 1.213 → 0.708). **Per-view depth metrics are worse**, though: val fine median 1.76 vs 1.52 mm, and scan48 fine median 3.96 vs 3.45 mm. Competing dips on scan48 fell from 18.75 to 12.90%.
- Fusion pass-through check: the confidence filter is close to a no-op, and the point counts are about the same as A0f's (22.3M vs 22.1M). At a pass rate matched to A0f, B2 still scores 0.416. So the gain comes from *where* the points land, not how many survive fusion. The exact mechanism has not been verified.

4. Open questions and next decisions

- **Is B2's gain robust?** Two of three seeds agree (0.414, 0.418), and the seed 1 result is due ~10-11. The decision on the 3-seed mean ± std comes from aggregate_b2_3seed.sh.
- **Does L_geo matter on B2?** Answered by the B2_nogeo seed 2 vs B2 seed 2 matched pair.
- **Can monocular information help without the noisy metric prior?** Answered by B2_mono_feat (frozen DA-V2-S features as a zero-init residual) vs B2_nogeo seed 2.
- **GeoCorr Lite decision (still open):** the evidence so far does not support the prior as the paper's main contribution. Capacity and monocular features are the remaining candidates.
- Why B2 fuses better despite worse per-view depth is not yet explained.

Sources: runs/*/eval_1600x1152.json, runs/A0f_fixes_only_short/aggregated_1600x1152.json, matched-pair diagnostics and probes under runs/, NOTES.md and the experiment-history doc. All numbers are measured; items not yet run are marked "not measured".